ISRO: India's Space Journey

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Introduction to ISRO

The Indian Space Research Organisation (ISRO) is the national space agency of India, headquartered in Bengaluru and functioning under the Department of Space of the Government of India. ISRO designs and launches rockets and satellites for communication, television broadcasting, weather forecasting, remote sensing, navigation and scientific exploration of the Moon, the Sun and Mars. It is known worldwide for its cost-effective missions and has become one of the leading space agencies of the world.

Founding and Early History

India's space programme began in 1962 when the Indian National Committee for Space Research (INCOSPAR) was set up under the Department of Atomic Energy at the initiative of Dr. Vikram Sarabhai and with the support of Prime Minister Jawaharlal Nehru. ISRO was formed on 15 August 1969, replacing INCOSPAR, and the Department of Space and the Space Commission were created in 1972. Dr. Vikram Ambalal Sarabhai is regarded as the founder, or father, of India's space programme. A physicist from Ahmedabad, he founded the Physical Research Laboratory and strongly believed that space technology should serve ordinary people in areas like education, communication and agriculture. The Thumba Equatorial Rocket Launching Station near Thiruvananthapuram in Kerala sent up India's first sounding rocket on 21 November 1963. Other leaders include Satish Dhawan, after whom the Sriharikota launch centre is named, U. R. Rao, K. Kasturirangan, K. Sivan, S. Somanath and, from January 2025, V. Narayanan.

India's First Satellite

India's first satellite was Aryabhata, named after the ancient Indian mathematician and astronomer. It was built by ISRO and launched on 19 April 1975 from the Soviet Union using a Soviet Kosmos-3M rocket. It carried instruments for X-ray astronomy, aeronomy and solar physics. Bhaskara-1, an Earth observation satellite, followed in 1979. The first Indian-built launch vehicle, SLV-3, successfully placed the Rohini satellite into orbit on 18 July 1980 from Sriharikota, with Dr. A. P. J. Abdul Kalam as the project director, making India the seventh country with satellite launch capability.

Launch Vehicles

ISRO has developed several launch vehicles. The Polar Satellite Launch Vehicle (PSLV), first successfully launched in 1994, is its reliable workhorse for putting satellites into polar orbits. In February 2017 PSLV-C37 launched 104 satellites in one mission, a world record at that time. The Geosynchronous Satellite Launch Vehicle (GSLV) carries heavier communication satellites to higher orbits and uses an indigenous cryogenic engine. The Launch Vehicle Mark-3 (LVM3), which is the most powerful ISRO rocket, launched Chandrayaan-3. The Small Satellite Launch Vehicle (SSLV) serves small satellites. The main spaceport is the Satish Dhawan Space Centre at Sriharikota in Andhra Pradesh, about 100 km north of Chennai. A new spaceport for small rockets is being built at Kulasekarapattinam in Thoothukudi district of Tamil Nadu.

Satellite Programmes

The INSAT series (Indian National Satellite System), started in 1983, is one of the largest domestic communication satellite systems in the Asia-Pacific region and supports telecommunication, television and weather observation. The Indian Remote Sensing (IRS) satellites, from 1988, provide images for agriculture, forestry, water resources and urban planning. The Cartosat series gives high-resolution images. NavIC is India's regional navigation system. The GSAT series carries communication payloads. Oceansat and the Megha-Tropiques mission support climate studies.

Major Space Missions

Chandrayaan-1 in October 2008 was India's first Moon mission. The Mars Orbiter Mission, nicknamed Mangalyaan, was launched on 5 November 2013 and entered Mars orbit on 24 September 2014, making India the first country to reach Mars orbit on its first attempt and the first in Asia, at a cost of about Rs. 450 crore. AstroSat, launched in September 2015, is India's first dedicated astronomy observatory. Chandrayaan-3 achieved a soft landing near the lunar south pole on 23 August 2023. Aditya-L1, launched on 2 September 2023, is India's first solar observatory and reached its halo orbit around the Sun-Earth L1 point on 6 January 2024. In December 2024, the SpaDeX mission demonstrated in-space docking, which India achieved in January 2025, making it the fourth country with this capability. In June 2025 Group Captain Shubhanshu Shukla became the first Indian astronaut to visit the International Space Station, on the private Axiom-4 mission. The NISAR satellite, a joint radar mission of NASA and ISRO, was launched in July 2025.

Gaganyaan and the Future

Gaganyaan is India's human spaceflight programme, which aims to send Indian astronauts to low-Earth orbit in an Indian spacecraft and bring them home safely, preceded by uncrewed test flights, including one planned with the humanoid robot Vyommitra. India also plans its own space station, the Bharatiya Antariksh Station, by about 2035, and an Indian crewed Moon landing by 2040. Chandrayaan-4, a lunar sample-return mission, and a Venus orbiter mission have been approved. The space sector was opened to private companies in 2020 through the regulator IN-SPACe, and the private space startups Skyroot Aerospace in Hyderabad and Agnikul Cosmos in Chennai have launched rockets. The commercial arm of ISRO is NewSpace India Limited (NSIL).

Benefits and Challenges

ISRO satellites help in cyclone warnings, such as for Cyclone Fani and Biparjoy, fisheries advisories, disaster management, tele-education and telemedicine, banking communications and navigation. Challenges include the need for heavier launch capacity, competition from private global players, funding and developing human-rated systems.

Quick Facts

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- Dr. Vikram Sarabhai is regarded as the father of India's space programme.
- India's first satellite was Aryabhata, launched on 19 April 1975.
- The Mars Orbiter Mission reached Mars orbit on 24 September 2014 on India's first attempt.
- PSLV-C37 launched 104 satellites in a single mission in February 2017.